FINAL PROJECT – FUNDAMENTALS OF DEEP LEARNING
MARCH 2025

PART 1 – FLICKR IMAGE CLASSIFICATION

Flickr is a website used for organizing, sharing and storing photos. It provides the tools for easy collaboration with one of the world's most passionate photo communities.[1] In this case, the company wants to implement a novel tool on the website to better categorize the images so the users can find what they need easily. For this task, the company hired you to create a classification model to label the images in categories such as: ***animals, objects, people, scenes***, and ***others***. For this purpose, you are provided with 8,000 images distributed in train and validation sets, and you have the following instructions to develop different CNN models.

This project will involve the following key steps:

1. Load the dataset and process the images (if required). Explain the steps you followed and decisions you made.
    - **a.** Train two different Multi-layer perceptron networks, explain the procedure and decisions taken to define the architectures. Show the number of parameters estimated and compare the results in terms of performance and architecture. (No. of layers, parameters, etc.)

2. Train a CNN with the structure of the diagram presented in the **APPENDIX.**
    - **a.** Define the number of epochs, batch size, and other parameters for training. Explain considering factors like convergence, computational resources, image size, number of samples, etc.
    - **b.** Train the model and compare the predictive performance on both the training and validation sets.
    - **c.** Discuss any observations, such as overfitting or underfitting, and potential steps for improvement.

3. Based on the previous results:
    - **a.** Conduct 3 experiments to improve the CNN based on the previous results (Consider processing limitations and accuracy performance). Explain the procedures and decisions taken to optimize the network settings, drawing on best practices discussed in class. Compare the loss and accuracy metrics across different models and between the training and validation sets. Justify any changes made to the architecture, hyperparameters, or training strategy in your pursuit of a better-performing model.

    - **b.** Implement data augmentation techniques to enhance your best model. Explain your procedure, specify if the preprocessing layers were applied directly to the dataset or included as part of the model. Retrain your best CNN with the augmented data and discuss the resulting improvements in performance and any observations regarding the model's accuracy and robustness.

c. Use two interpretability techniques to explain the decisions of your best performing model. You can include techniques not covered during the class example. Reference **https://arxiv.org/pdf/2004.14545.pdf**
d. Use transfer learning to create a new model. Explain the reasons why you chose a certain model and the adjustments you have made. (You can use one of the pretrained models from keras: https://keras.io/api/applications/)
e. Make a conclusion comparing your results (MLP, CNN's and other variations).

PART 2 - EXPLORING RNNS & CNNS FOR A SEQUENCE PREDICTION TASK

This project delves into the application of two prominent deep learning architectures: Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs). We aim to explore their suitability for a specific sequence prediction task using a chosen dataset.

a. **Dataset and RNN Selection:** Find a Dataset where you can apply RNN's.
Dataset: Specify the chosen dataset, with details of **description**, **source**, **task category**, and explain **why it is appropriate to apply RNN's** in this case
Note: Task Type falls under the category of many-to-many, one-to-many, etc.

b. **RNN Experimentation and Attention:** Conduct experiments by training and evaluating various RNN architectures, including LSTMs and GRUs, with different hyperparameter configurations (e.g., number of layers, units per layer). Apply attention and memory to the best performing model. Compare the results and show how the attention mechanism is working (examples).

c. **CNN Comparison:** Develop and train a relevant CNN architecture appropriate for the same dataset. Compare the performance of the best-performing RNN model with the CNN and Explain the differences in the architecture.

Deliverables:

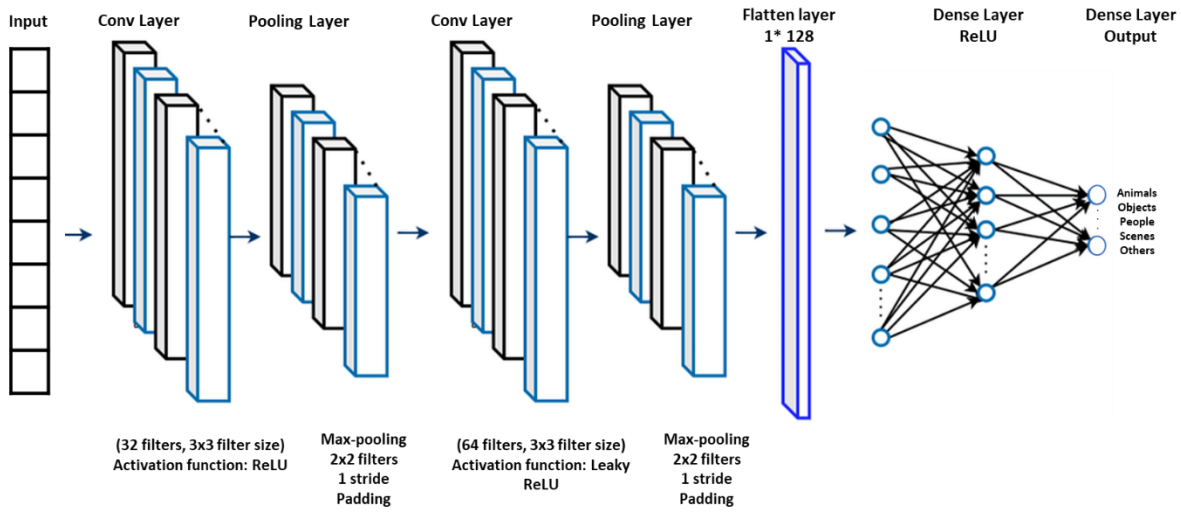
- Notebook including all the code to support your results and conclusions.
    - You should provide all the code required to be able to replicate the experiments and sufficient documentation to understand the steps and choices you made.
- A detailed report:
    - Include plots, examples and all the information you consider relevant to support your results.
    - Describe the steps and motivation of the choices you made in developing the various models.
    - There is no page limit to your report, but it should be both concise and sufficient.

Hand-in: 1ST APRIL 2025 - .zip folder containing the notebook and report (PDF) ***Don't forget to include all the sources and references used during the project. Follow the IESEG ANTI-PLAGIARISM POLICY - An adapted version for an Age of Generative Artificial Intelligence***

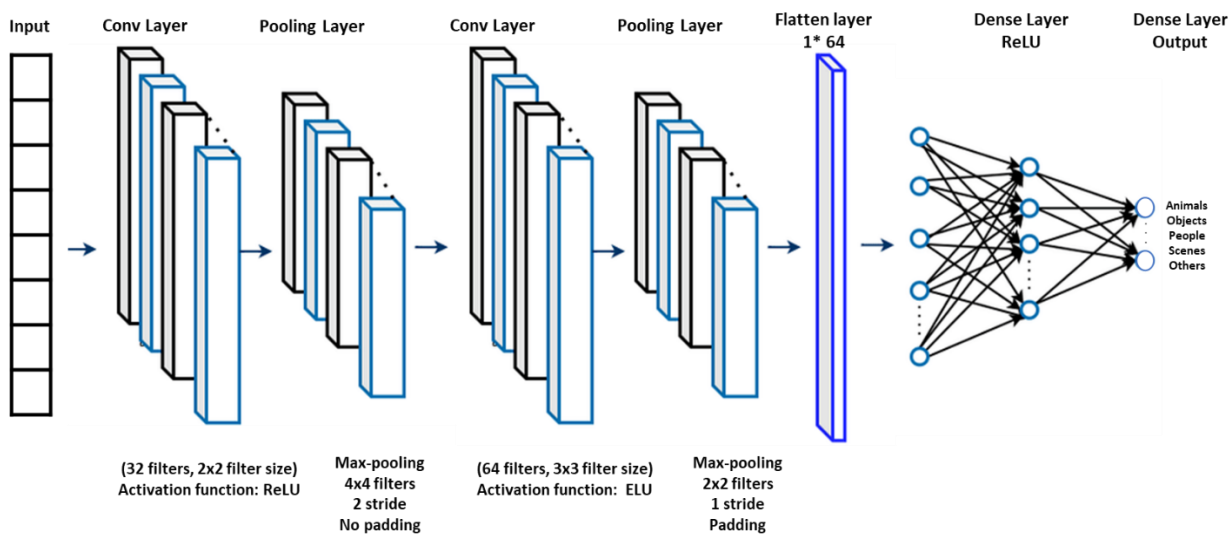
APPENDIX

Use the diagram corresponding to your group number:

Group 1:



Group 2:



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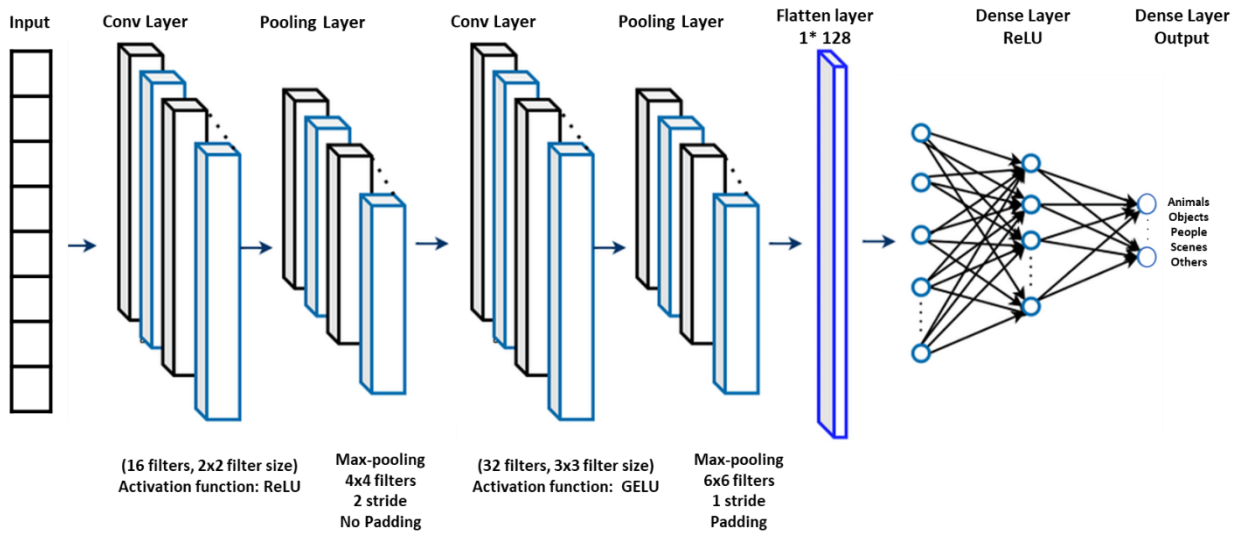
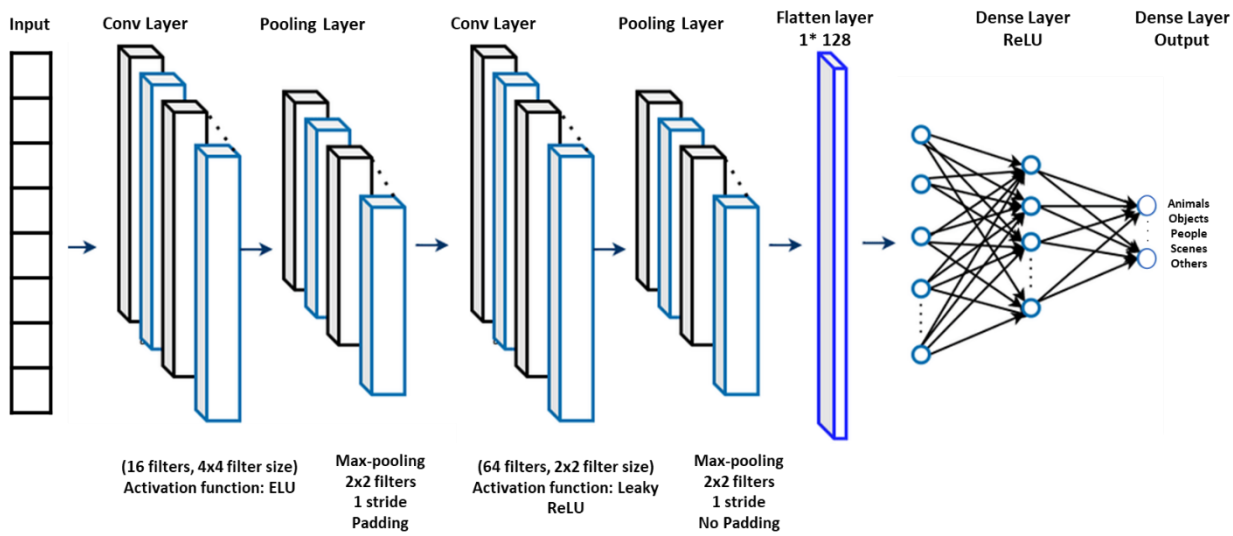
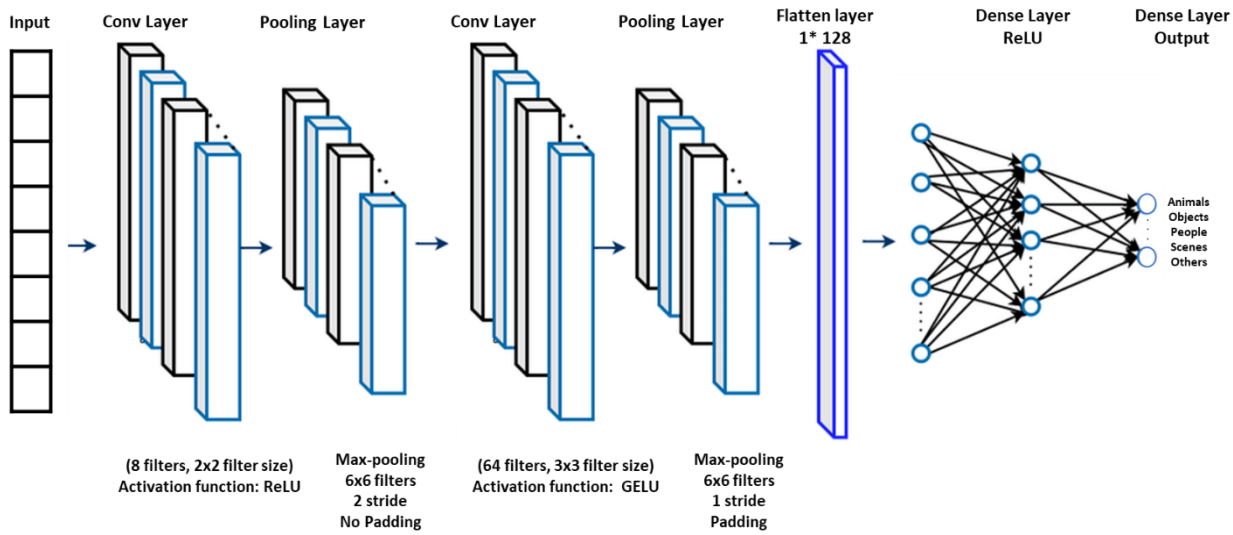


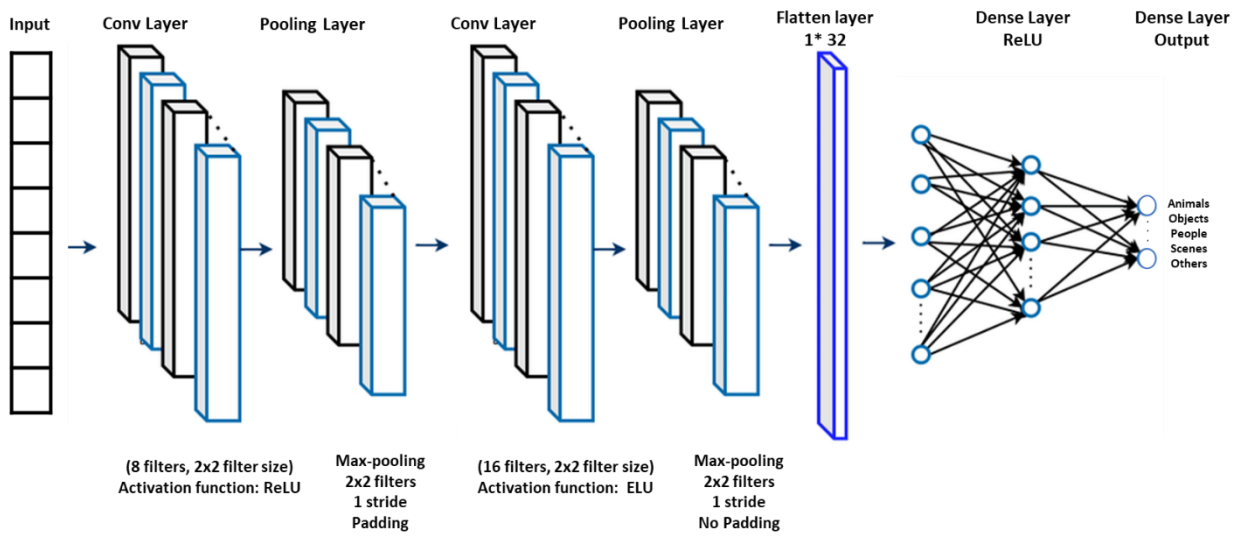
Group 4:



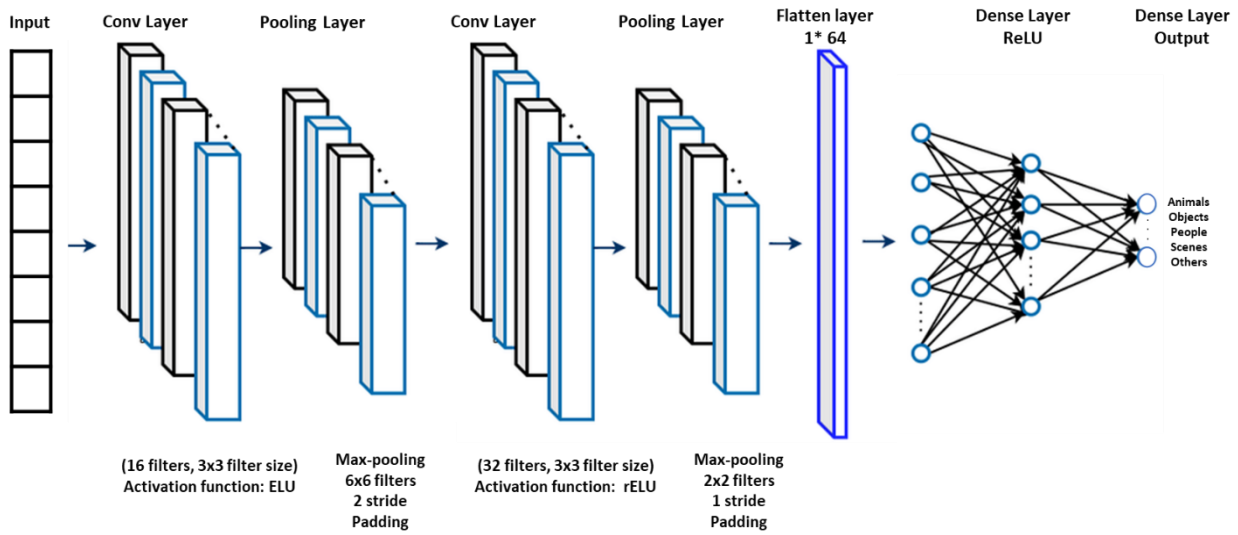
Group 5:



Group 6:



Group 7:



Diagrams based on: *https://www.researchgate.net/figure/The-single-CNN-architecture-The-network-includes-following-layers-input-layer-one_fig1_358477638*